Data Communication & Computer Networks (CN)

Topic 2 of 7 : Network Models (OSI & TCP/IP)

81 MCQs — Repo Exam Questions + Forouzan — bilingual (English + Hindi), exam-ready

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)**

Total Questions: **81** MCQs + **20** Nukte rapid-fire revision points

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

Syllabus Coverage — sub-topic wise order

1. Section A — OSI Model: Basics, Layer Order & Support Layers — Q1 - Q12

2. Section B — OSI Layer Functions (Layer 1 → Layer 7) — Q13 - Q63

3. Section C — TCP/IP Model — Q64 - Q74

4. Section D — Comparison, Encapsulation & PDUs — Q75 - Q81

How to read: Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

System: generated by **tools/cn_build_topic.py** from **tools/data/topic2_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

Section A — OSI Model: Basics, Layer Order & Support Layers

OSI = Open Systems Interconnection (ISO, 1984): 7 layers, layer order, network vs user support layers, address types.

Q1. OSI stands for:
OSI का पूरा नाम (full form) क्या है?

(A) Open Systems Interconnection **(B)** Operating System Interface
(C) Optical Service Implementation **(D)** None of these

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) Open Systems Interconnection** — OSI = Open Systems Interconnection — the reference model published by ISO in 1984 that standardises network communication into 7 layers.
OSI = Open Systems Interconnection — ISO द्वारा 1984 में publish किया गया reference model जो network communication को 7 layers में बाँटता है।
★ **Nukta:** OSI = Open Systems Interconnection (ISO, 1984, 7 layers). Ye sabse favourite one-liner hai.
OSI = Open Systems Interconnection (ISO, 1984, 7 layers)। सबसे favourite one-liner।

Q2. The OSI model was developed by:
OSI model किसने develop किया?

(A) IEEE **(B)** ISO
(C) IETF **(D)** ITU

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) ISO** — ISO (International Organization for Standardization) published the OSI reference model in 1984.
ISO (International Organization for Standardization) ने OSI reference model 1984 में publish किया।
★ **Nukta:** OSI = ISO ka (1984). TCP/IP = DoD/ARPA ka. IEEE = 802 LAN standards. IETF = TCP/IP standards.
OSI = ISO का (1984)। TCP/IP = DoD/ARPA का। IEEE = 802 LAN standards। IETF = TCP/IP standards।

Q3. The OSI model has how many layers?
OSI model में कितनी layers हैं?

(A) 5 **(B)** 6
(C) 7 **(D)** 8

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (C) 7** — The OSI reference model has exactly 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.
OSI reference model में ठीक 7 layers हैं: Physical, Data Link, Network, Transport, Session, Presentation, Application।
★ **Nukta:** Mnemonic: Please Do Not Throw Sausage Pizza Away (Physical, Data-link, Network, Transport, Session, Presentation, Application).
Mnemonic: Please Do Not Throw Sausage Pizza Away (Physical, Data-link, Network, Transport, Session, Presentation, Application)।

Q4. Why was the OSI model developed?
OSI model क्यों develop किया गया?

(A) Manufacturers disliked the TCP/IP protocol suite / Manufacturers को TCP/IP protocol suite पसंद नहीं था **(B)** The rate of data transfer was increasing exponentially / Data transfer की दर तेज़ी से बढ़ रही थी
(C) Standards were needed to allow any two systems to communicate / किन्हीं भी दो systems के बीच communication के लिए standards चाहिए थे **(D)** None of the above / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (C) Standards were needed to allow any two systems to communicate** — OSI was developed to provide an international standard so that any two systems from different vendors could communicate (open systems interconnection).
OSI एक international standard देने के लिए बना ताकि अलग vendors के कोई भी दो systems आपस में communicate कर सकें।
★ **Nukta:** OSI = OPEN systems — matlab koi bhi do systems jud sakein. Isliye 'standards were needed' sahi jawab hai.
OSI = OPEN systems — कोई भी दो systems जुड़ सकें। इसलिए 'standards were needed' सही उत्तर है।

Q5. In the OSI model, when data is sent from device A to device B, which is the 5th layer to receive data at B (counting from the bottom)?

OSI model में जब data device A से device B को भेजा जाता है, तो B पर data पाने वाली 5वीं layer (नीचे से गिनती में) कौन-सी है?

(A) Application layer **(B)** Transport layer
(C) Link layer **(D)** Session layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (D) Session layer** — Counting up at the receiver: 1 Physical, 2 Data Link, 3 Network, 4 Transport, 5 Session. So the 5th layer to receive is the Session layer.

Receiver पर गिनती: 1 Physical, 2 Data Link, 3 Network, 4 Transport, 5 Session। इसलिए 5वीं layer Session है।

★ **Nukta:** Layer order bottom-up: Physical(1), Data-link(2), Network(3), Transport(4), Session(5), Presentation(6), Application(7).

Layer क्रम (bottom-up): Physical(1), Data-link(2), Network(3), Transport(4), Session(5), Presentation(6), Application(7)।

Q6. The session, presentation, and application layers are the _____ support layers.

Session, Presentation और Application layers _____ support layers हैं।

(A) user **(B)** network
(C) both (a) and (b) **(D)** neither (a) nor (b)

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) user** — Layers 5-6-7 (Session, Presentation, Application) are called USER support layers; layers 1-2-3 are NETWORK support layers; Transport (layer 4) joins both.

Layers 5-6-7 (Session, Presentation, Application) USER support layers कहलाती हैं; layers 1-2-3 NETWORK support layers; Transport (layer 4) दोनों को जोड़ती है।

★ **Nukta:** User support = 5,6,7. Network support = 1,2,3. Transport(4) = dono ka bridge.

User support = 5,6,7। Network support = 1,2,3। Transport(4) = दोनों का bridge।

Q7. Which of the following is NOT a network support layer?

निम्नलिखित में से कौन network support layer नहीं है?

(A) Transport Layer **(B)** Network Layer
(C) Data Link Layer **(D)** Physical Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) Transport Layer** — Network support layers = Physical(1), Data Link(2), Network(3). Transport(4) is the bridge between network and user support layers, so it is not counted as a network support layer.

Network support layers = Physical(1), Data Link(2), Network(3)। Transport(4) network और user support layers के बीच bridge है।

★ **Nukta:** Yaad rakho: NOT-network-support = Transport. (1,2,3 = network support; 5,6,7 = user support.)

याद रखो: NOT-network-support = Transport। (1,2,3 = network support; 5,6,7 = user support.)

Q8. Network support layers are:

Network support layers कौन-सी हैं?

(A) 1, 2, 3 **(B)** 4, 5
(C) 5, 6, 7 **(D)** 4 only

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) 1, 2, 3** — Physical(1), Data Link(2) and Network(3) form the network support layers — they move data between the two end systems.

Physical(1), Data Link(2) और Network(3) मिलकर network support layers बनाती हैं।

★ **Nukta:** 1,2,3 = network support; 5,6,7 = user support; 4 = bridge. Exam mein ye grouping bar-bar aata hai.

1,2,3 = network support; 5,6,7 = user support; 4 = bridge।

Q9. Which layer connects the network support layers and the user support layers?

कौन-सी layer network support layers और user support layers को जोड़ती है?

(A) Transport layer **(B)** Network layer
(C) Data Link layer **(D)** Session layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) Transport layer** — Transport (layer 4) sits between the network support layers (1-3) and user support layers (5-7) and joins them — it is often called the bridge layer.

Transport (layer 4) network support layers (1-3) और user support layers (5-7) के बीच बैठकर दोनों को जोड़ती है — इसे bridge layer कहते हैं।

★ **Nukta:** Transport = bridge layer (na upar wale, na neeche wale — dono se judta hai).

Transport = bridge layer (ऊपर और नीचे दोनों से जुड़ती है)।

Q10. In the ISO-OSI reference model, the Session layer offers dialog control, token management and _____ as services.

ISO-OSI reference model में Session layer dialog control, token management और _____ services देती है।

(A) Synchronization **(B)** Asynchronization
(C) Errors **(D)** Flow control

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (A) Synchronization** — The Session layer's three classic services are dialog control, token management and synchronisation (checkpointing for recovery).

Session layer की तीन classic services हैं: dialog control, token management और synchronization (recovery के लिए checkpointing)।

★ **Nukta:** Session ki 3 services: Dialog control + Token management + Synchronization (checkpoint).

Session की 3 services: Dialog control + Token management + Synchronization (checkpoint)।

Q11. How many address types (identifiers) are used from sender to receiver in OSI/TCP-IP communication?

OSI/TCP-IP communication में sender से receiver तक कितने address types (identifiers) use होते हैं?

(A) 2 **(B)** 3
(C) 4 **(D)** 5

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (C) 4** — Forouzan: four levels of addresses are used — physical (link) address, logical (IP) address, port address and specific (user-friendly) address.

Forouzan: चार levels के addresses use होते हैं — physical (link) address, logical (IP) address, port address और specific (user-friendly) address।

★ **Nukta:** 4 addresses: Physical(48-bit) / IP(32-bit) / Port(16-bit) / Specific (user-friendly).

4 addresses: Physical(48-bit) / IP(32-bit) / Port(16-bit) / Specific।

Q12. Who developed the standards for the OSI reference model?

OSI reference model के standards किसने develop किए?

(A) ANSI - American National Standards Institute **(B)** ISO - International Organization for Standardization
(C) IEEE - Institute of Electrical and Electronics Engineers **(D)** ACM - Association for Computing Machinery

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (B) ISO - International Organization for Standardization** — ISO (International Organization for Standardization) developed and published the OSI reference model standards.

ISO (International Organization for Standardization) ने OSI reference model standards develop और publish किए।

★ **Nukta:** OSI ke standards = ISO ke. ANSI/IEEE/ACM ke alag kaam hain (US standards / LAN / computing society).

OSI के standards = ISO के। ANSI/IEEE/ACM के अलग काम हैं।

Section B — OSI Layer Functions (Layer 1 → Layer 7)

Which layer does what: Physical, Data Link (LLC/MAC, framing, FCS), Network (logical addressing/routing), Transport (segmentation, end-to-end, ports), Session (dialog/checkpoint), Presentation (translation/encryption/compression), Application (services & protocols).

Q13. Fiber optic cables and coaxial cables function at which OSI layer?

Fiber optic cables और coaxial cables किस OSI layer पर काम करती हैं?

(A) Data Link Layer **(B)** Physical Layer
(C) Network Layer **(D)** Transport Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (B) Physical Layer** — Transmission media (fiber, coaxial, twisted pair) are used at the Physical layer — it transmits raw bits over the medium.

Transmission media (fiber, coaxial, twisted pair) Physical layer पर use होती हैं — यह medium पर raw bits भेजती है।

★ **Nukta:** Media (cable, fiber, radio) = Physical layer ka kaam.

Media (cable, fiber, radio) = Physical layer का काम।

Q14. The term 'Baud Rate' is associated with which layer?

'Baud Rate' शब्द किस layer से जुड़ा है?

(A) Network Layer **(B)** Data Link Layer
(C) Physical Layer **(D)** Application Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (C) Physical Layer** — Baud rate = number of signal changes per second — a Physical layer concept (signal/line encoding).

Baud rate = प्रति सेकंड signal changes की संख्या — Physical layer की concept (signal/line encoding)।

★ **Nukta:** Baud rate = signalling rate = Physical layer. Bit rate ≠ baud rate (bits per symbol yaad rakho).

Baud rate = signalling rate = Physical layer। Bit rate ≠ baud rate।

Q15. The layer used to deal with mechanical and electrical specifications is:

Mechanical और electrical specifications से निपटने वाली layer कौन-सी है?

(A) Physical Layer **(B)** Data Link Layer
(C) Network Layer **(D)** Transport Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) Physical Layer** — The Physical layer defines mechanical, electrical, functional and procedural specifications of the physical link (connectors, voltages, pin layout, signals).

Physical layer physical link की mechanical, electrical, functional और procedural specifications define करती है (connectors, voltages, pin layout, signals)।

★ **Nukta:** Mechanical + Electrical + Functional + Procedural = Physical layer ke 4 specifications.

Mechanical + Electrical + Functional + Procedural = Physical layer की 4 specifications।

Q16. The Physical layer deals with:

Physical layer किससे निपटती है?

(A) Logical addressing **(B)** Media, signals and binary transmission / Media, signals और binary transmission
(C) Error correction **(D)** Segmentation

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (B) Media, signals and binary transmission** — Physical layer = media, signals, line coding and bit (binary) transmission. Logical addressing = Network; error correction = Data Link/Transport; segmentation = Transport.

Physical layer = media, signals, line coding और bit transmission। Logical addressing = Network; error correction = Data Link/Transport; segmentation = Transport।

★ **Nukta:** Physical = bits + signals + media. Baaki sab upar ki layers ka kaam hai.

Physical = bits + signals + media। बाकी ऊपर की layers का काम है।

Q17. The Physical layer is responsible for which of the following operations?

Physical layer निम्नलिखित में से किन operations के लिए जिम्मेदार है?

(A) Line coding **(B)** Channel coding
(C) Modulation **(D)** All of the above

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (D) All of the above** — Line coding, channel coding and modulation are all Physical layer functions — they prepare the bits for transmission over the medium.

Line coding, channel coding और modulation — तीनों Physical layer के functions हैं।

★ **Nukta:** Physical layer = Line coding + Channel coding + Modulation + signals/media — 'All of the above' yaad rakho.

Physical layer = Line coding + Channel coding + Modulation। 'All of the above' याद रखो।

Q18. Transmission data rate is decided by:

Transmission data rate कौन तय करती है?

(A) Network layer **(B)** Physical layer
(C) Data Link layer **(D)** Transport layer

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)]

✔ **Correct: (B) Physical layer** — The data (bit) rate is a Physical layer property — it depends on the bandwidth of the medium and the encoding/modulation used.

Data (bit) rate Physical layer की property है — यह medium की bandwidth और encoding/modulation पर निर्भर करती है।

★ **Nukta:** Data rate = Physical layer. (Throughput/delay performance bhi isi layer ke signals par depend karti hai.)

Data rate = Physical layer।

Q19. Repeaters function in:

Repeaters किस layer पर काम करते हैं?

(A) Physical layer **(B)** Data link layer
(C) Network layer **(D)** Both (A) and (B)

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (A) Physical layer** — A repeater regenerates the electrical/optical signal — a Physical layer device (layer 1).

Repeater electrical/optical signal regenerate करता है — Physical layer device (layer 1)।

★ **Nukta:** Repeater/Hub = Physical (L1). Bridge/Switch = Data Link (L2). Router = Network (L3).

Repeater/Hub = Physical (L1)। Bridge/Switch = Data Link (L2)। Router = Network (L3)।

Q20. The physical layer is responsible for:
Physical layer किसके लिए जिम्मेदार है?

(A) Bit-by-bit delivery **(B)** Port to port delivery
(C) Process to process delivery **(D)** Application to application delivery

[Source: RPSC Programmer 2024 Paper-1 (with official key)]

✔ **Correct: (A) Bit-by-bit delivery** — Physical layer = transmission of individual bits over the link (bit-by-bit delivery). Port-to-port = Transport; process-to-process = Transport; application-to-application = Application. (Matches the official RPSC key.)

Physical layer = link पर bit-by-bit delivery। Port-to-port = Transport; process-to-process = Transport; application-to-application = Application। (Official RPSC key से match.)

★ **Nukta:** Bit-by-bit = Physical. Hop-to-hop = Data Link. Source-to-destination = Network. Process-to-process = Transport.

Bit-by-bit = Physical। Hop-to-hop = Data Link। Source-to-destination = Network। Process-to-process = Transport।

Q21. The sub-layer of the Physical layer which directly interfaces with the MAC sub-layer is called:
Physical layer का वह sub-layer जो सीधे MAC sub-layer से जुड़ता है, कहलाता है:

(A) Physical signalling sub-layer **(B)** Physical data sub-layer
(C) Physical address sub-layer **(D)** Physical transport sub-layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (A) Physical signalling sub-layer** — In the IEEE 802 architecture the Physical Layer Signalling (PLS) sub-layer sits directly under the MAC sub-layer and passes frames to/from it.

IEEE 802 architecture में Physical Layer Signalling (PLS) sub-layer सीधे MAC sub-layer के नीचे होता है और frames आदान-प्रदान करता है।

★ **Nukta:** PLS (Physical Layer Signalling) = Physical layer ka upper sub-layer jo MAC se directly judta hai.

PLS (Physical Layer Signalling) = Physical layer का upper sub-layer जो MAC से जुड़ता है।

Q22. The physical layer translates logical communication requests from the _____ into hardware-specific operations.
Physical layer, _____ के logical communication requests को hardware-specific operations में बदलता है।

(A) Network layer **(B)** Transport layer
(C) Data link layer **(D)** Application layer

[Source: InstrumentationTools 150 CN MCQs (key corrected vs Forouzan)]

✔ **Correct: (C) Data link layer** — The data link layer (directly above the physical layer) sends logical requests like 'transmit this frame'; the physical layer converts them into the electrical/optical hardware operations on the medium.

Data link layer (physical layer के ठीक ऊपर) जैसे logical requests भेजती है — 'यह frame transmit करो'; physical layer उन्हें medium पर electrical/optical hardware operations में बदलती है।

★ **Nukta:** Each layer serves the layer ABOVE it — physical serves data link. Exam banks sometimes mis-key this as 'network layer'; the correct neighbour is the data link layer.

हर layer अपने ऊपर वाली layer की सेवा करती है — physical, data link की। कुछ exam banks इसे गलती से 'network layer' लिखते हैं; सही उत्तर data link layer है।

Q23. Which one of the following is a data link (Layer 2) protocol?
निम्न में से कौन data link (Layer 2) protocol है?

(A) Point-to-Point Protocol (PPP) **(B)** HDLC
(C) Ethernet **(D)** All of these

[Source: InstrumentationTools 150 CN MCQs]

✔ **Correct: (D) All of these** — PPP, HDLC and Ethernet are all data link layer (Layer 2) protocols — they frame packets and handle physical addressing/error detection at the link level.

PPP, HDLC और Ethernet — तीनों data link layer (Layer 2) protocols हैं — ये packets को frame करते हैं और link level पर physical addressing/error detection करते हैं।

★ **Nukta:** Layer 2 protocols = PPP, HDLC, Ethernet, Frame Relay, ATM (cell-based). Network layer = IP, ICMP, ARP.

Layer 2 protocols = PPP, HDLC, Ethernet, Frame Relay, ATM (cell-based)। Network layer = IP, ICMP, ARP।

Q24. The Application layer is implemented in:
Application layer कहाँ implement होती है?
(A) End system **(B)** NIC
(C) Ethernet **(D)** Packet transport
[Source: Sanfoundry CN MCQs — Reference Models]

✔ **Correct: (A) End system** — In the protocol stack, the application layer (like transport) is implemented in the end systems — the hosts (user's computer or the server) — not inside intermediate network devices.
Protocol stack में application layer (transport की तरह) end systems में implement होती है — hosts (user का computer या server) में — बीच के network devices में नहीं।
★ **Nukta:** End systems (hosts) implement layers up to the application; switches implement only Layers 1-2, routers only Layers 1-3.
End systems (hosts) application तक की layers implement करते हैं; switches सिर्फ Layers 1-2, routers सिर्फ Layers 1-3।

Q25. Which layer of the OSI model is responsible for the transmission and reception of unstructured raw data between a device and a physical transmission medium?
OSI model की कौन-सी layer device और physical transmission medium के बीच unstructured raw data के transmission और reception के लिए जिम्मेदार है?
(A) Transport layer **(B)** Data-Link layer
(C) Network layer **(D)** Physical layer
[Source: Central exams — SSC/Railway/UPSC Computer Awareness (Exampundit)]

✔ **Correct: (D) Physical layer** — Raw (unstructured) bit transmission over the medium is exactly the Physical layer's role.
Medium पर raw (unstructured) bits भेजना Physical layer का ही काम है।
★ **Nukta:** 'Raw data + physical medium' = Physical layer. SSC/Railway mein ye one-liner common hai.
'Raw data + physical medium' = Physical layer।

Q26. Which layer creates the 'collision domain'?
'Collision domain' कौन-सी layer बनाती है?
(A) Physical Layer (hubs) **(B)** Data Link Layer (switches/bridges)
(C) Network Layer **(D)** Transport Layer
[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (A) Physical Layer (hubs)** — A collision domain exists on a shared medium — the shared physical connection created by hubs/repeaters (Physical layer devices). Switches/bridges (Data Link) SEPARATE collision domains; routers separate broadcast domains.
Collision domain shared medium पर बनता है — hubs/repeaters (Physical layer devices) के shared connection से। Switches/bridges (Data Link) collision domains को अलग करते हैं; routers broadcast domains अलग करते हैं।
★ **Nukta:** Hub network = ONE collision domain (Physical). Switch = per-port collision domain (Data Link). Router = broadcast domain boundary (Network).
Hub network = एक collision domain (Physical)। Switch = per-port collision domain (Data Link)। Router = broadcast domain boundary (Network)।

Q27. Data Link Layer is divided into:
Data Link Layer किनमें विभाजित है?
(A) LLC and MAC **(B)** Physical and Network
(C) Transport and Session **(D)** Application and Presentation
[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (A) LLC and MAC** — The IEEE 802 Data Link layer has two sub-layers: LLC (Logical Link Control, 802.2) and MAC (Media Access Control, e.g. 802.3).
IEEE 802 Data Link layer के दो sub-layers हैं: LLC (Logical Link Control, 802.2) और MAC (Media Access Control, जैसे 802.3)।
★ **Nukta:** Data Link = LLC (upper, 802.2) + MAC (lower, 802.3). Ye hamesha saath aate hain.
Data Link = LLC (ऊपर, 802.2) + MAC (नीचे, 802.3)।

Q28. Which statement best describes the difference between the LLC and MAC sub-layers?

LLC और MAC sub-layers के बीच अंतर सबसे अच्छा कौन बताता है?

(A) LLC handles physical addressing, MAC handles flow control / LLC physical addressing करता है, MAC flow control करता है **(B)** LLC provides the service interface to the Network layer; MAC controls access to the medium / LLC, Network layer को service interface देता है; MAC medium तक access control करता है

(C) LLC works only in wireless networks, MAC only in wired networks / LLC सिर्फ wireless में, MAC सिर्फ wired में काम करता है **(D)** LLC performs framing, MAC performs routing / LLC framing करता है, MAC routing करता है

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)]

✔ **Correct: (B) LLC provides the service interface to the Network layer; MAC controls access to the medium** — LLC (802.2) provides an interface to the Network layer (flow/error control over the link); MAC (802.3 etc.) handles physical addressing and media access (CSMA/CD etc.).

LLC (802.2) Network layer को interface देता है; MAC (802.3 आदि) physical addressing और media access (CSMA/CD आदि) संभालता है।

★ **Nukta:** LLC = upar wala sub-layer (network layer se link). MAC = neeche wala (medium access + physical address).

LLC = ऊपर का sub-layer। MAC = नीचे का (medium access + physical address)।

Q29. Physical (hardware) addressing is added at:

Physical (hardware) addressing किस layer पर जोड़ी जाती है?

(A) Network **(B)** Transport

(C) Data Link **(D)** Session

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (C) Data Link** — The MAC/physical address (48-bit) is added to the frame header at the Data Link layer.

MAC/physical address (48-bit) frame header में Data Link layer पर जोड़ी जाती है।

★ **Nukta:** Physical/MAC address = Data Link. IP/Logical = Network. Port = Transport.

Physical/MAC address = Data Link। IP/Logical = Network। Port = Transport।

Q30. A frame is received at the Data Link layer with an incorrect FCS. What will happen?

Data Link layer पर frame गलत FCS के साथ प्राप्त होता है। क्या होगा?

(A) Frame discarded **(B)** Frame corrected

(C) Sent to network layer **(D)** Re-transmitted automatically

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (A) Frame discarded** — FCS (Frame Check Sequence, usually CRC) detects errors; a frame with a bad FCS is silently discarded by the receiver's Data Link layer (retransmission, if any, is a higher-level/ARQ decision).

FCS (Frame Check Sequence, सामान्यतः CRC) errors detect करता है; गलत FCS वाला frame receiver की Data Link layer द्वारा discard कर दिया जाता है।

★ **Nukta:** Bad FCS = frame drop (discard). Correction/ Data Link khud re-transmit nahi karti — wo ARQ/higher layer ka decision hai.

गलत FCS = frame discard।

Q31. 'Transparent transmission' is the function of which layer?

'Transparent transmission' किस layer का काम है?

(A) Data link **(B)** Network

(C) Session **(D)** Transport

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (A) Data link** — The Data Link layer makes the physical link look error-free/transparent to the Network layer — it hides all bit-level details.

Data Link layer physical link को Network layer के लिए error-free/transparent बनाती है — सारी bit-level details छिपा देती है।

★ **Nukta:** Data Link = 'transparent service' — Network layer ko pata hi nahi chalta ki neeche kitni mehnat hui.

Data Link = 'transparent service' — Network layer को नीचे की मेहनत पता नहीं चलती।

Q32. Which layer handles 'hop-to-hop' flow control?

'Hop-to-hop' flow control कौन-सी layer संभालती है?

(A) Network **(B)** Data Link

(C) Transport **(D)** Physical

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) Data Link** — Hop-to-hop (node-to-node) flow control is done by the Data Link layer between two adjacent nodes; end-to-end flow control is Transport's job.

Hop-to-hop (node-to-node) flow control Data Link layer करती है; end-to-end flow control Transport का काम है।

★ **Nukta:** Hop-to-hop = Data Link. End-to-end = Transport. Dono flow control karte hain — scope alag hai.

Hop-to-hop = Data Link। End-to-end = Transport।

Q33. The _____ layer is responsible for delivering data units from one station to the next without errors.

_____ layer data units को एक station से अगले station तक बिना errors पहुँचाने के लिए जिम्मेदार है।

(A) Transport **(B)** Network
(C) Data link **(D)** Physical

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (C) Data link** — Forouzan: the Data Link layer is responsible for delivering data units from one station to the NEXT without errors (hop-to-hop delivery).

Forouzan: Data Link layer data units को एक station से अगले तक बिना errors पहुँचाती है (hop-to-hop delivery)।

★ **Nukta:** 'Station to next station' = Data Link (hop-to-hop). 'Source to destination' = Network.

'Station to next station' = Data Link। 'Source to destination' = Network।

Q34. _____ Layer is responsible for framing, error detection and physical addressing.

_____ Layer framing, error detection और physical addressing के लिए जिम्मेदार है।

(A) Network **(B)** Transport
(C) Application **(D)** Data Link

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (D) Data Link** — Framing, error detection (CRC/FCS) and physical (MAC) addressing — all three are classic Data Link layer responsibilities.

Framing, error detection (CRC/FCS) और physical (MAC) addressing — तीनों Data Link layer की जिम्मेदारी हैं।

★ **Nukta:** Framing + Error detection + MAC addressing = Data Link ka 'trio'.

Framing + Error detection + MAC addressing = Data Link का trio।

Q35. Bridge works in which layer of the OSI model?

Bridge OSI model की किस layer पर काम करता है?

(A) Application layer **(B)** Transport layer
(C) Network layer **(D)** Data link layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2) · also asked in KVS PGT CS 2018 Official Paper]

✔ **Correct: (D) Data link layer** — A bridge connects two LAN segments and filters/forwards frames using MAC addresses — a Data Link layer (layer 2) device.

Bridge दो LAN segments को जोड़ता है और MAC addresses से frames filter/forward करता है — Data Link layer (layer 2) device।

★ **Nukta:** Bridge/Switch = L2. Router = L3. Hub/Repeater = L1. Gateway = L4+.

Bridge/Switch = L2। Router = L3। Hub/Repeater = L1। Gateway = L4+।

Q36. From which layer does the Data Link layer collect packets and encapsulate them into frames for transmission?

Data Link layer किस layer से packets लेकर उन्हें transmission के लिए frames में encapsulate करती है?

(A) Transport layer **(B)** Application layer
(C) Network layer **(D)** Physical layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (C) Network layer** — The Data Link layer receives packets (datagrams) from the Network layer above and encapsulates them into frames.

Data Link layer ऊपर की Network layer से packets (datagrams) लेती है और उन्हें frames में encapsulate करती है।

★ **Nukta:** Encapsulation: Network packet -> Data Link frame -> Physical bits. Upar se neeche.

Encapsulation: Network packet -> Data Link frame -> Physical bits।

Q37. Which OSI layer is responsible for logical addressing?

Logical addressing के लिए कौन-सी OSI layer जिम्मेदार है?

(A) Data Link **(B)** Network
(C) Transport **(D)** Application

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) Network** — Logical (IP) addressing and routing are Network layer functions (layer 3).

Logical (IP) addressing और routing Network layer (layer 3) के functions हैं।

★ **Nukta:** Logical address = IP = Network layer (L3). Physical address = MAC = Data Link (L2).

Logical address = IP = Network layer (L3)। Physical address = MAC = Data Link (L2)।

Q38. IPv4 and IPv6 work at which OSI layer?

IPv4 और IPv6 किस OSI layer पर काम करते हैं?

(A) Data link layer **(B)** Network layer
(C) Transport layer **(D)** Application layer

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)]

✔ **Correct: (B) Network layer** — IP (IPv4/IPv6) is the Network layer protocol of the Internet — layer 3.

IP (IPv4/IPv6) Internet का Network layer protocol है — layer 3।

★ **Nukta:** IP = Internet Protocol = Network/Internet layer (L3). Hamesha!

IP = Network/Internet layer (L3)।

Q39. The _____ layer is responsible for the source-to-destination delivery of a packet across multiple network links.

_____ layer एक packet की multiple network links के बीच source-to-destination delivery के लिए जिम्मेदार है।

(A) Transport **(B)** Network
(C) Data link **(D)** Physical

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (B) Network** — Forouzan: the Network layer handles source-to-destination (host-to-host) delivery of a packet across multiple links via routing.

Forouzan: Network layer routing के जरिए packet की multiple links के बीच source-to-destination delivery करती है।

★ **Nukta:** Source-to-destination = Network. Process-to-process = Transport. Station-to-station = Data Link.

Source-to-destination = Network। Process-to-process = Transport। Station-to-station = Data Link।

Q40. Which layer performs segmentation?

Segmentation कौन-सी layer करती है?

(A) Data link **(B)** Transport
(C) Network **(D)** Session

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) Transport** — The Transport layer splits the message into segments and reassembles them at the destination (segmentation and reassembly).

Transport layer message को segments में बाँटती है और destination पर जोड़ती है (segmentation and reassembly)।

★ **Nukta:** Segmentation + Reassembly = Transport. (Packetisation Network ki, Framing Data Link ki.)

Segmentation + Reassembly = Transport। (Packetisation Network की, Framing Data Link की।)

Q41. Which layer provides end-to-end communication?

End-to-end communication कौन-सी layer देती है?

(A) Network **(B)** Transport
(C) Session **(D)** Physical

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) Transport** — Transport provides end-to-end (process-to-process) communication between the two end hosts.

Transport दोनों end hosts के बीच end-to-end (process-to-process) communication देती है।

★ **Nukta:** End-to-end = Transport. Source-to-destination = Network. Hop-to-hop = Data Link. Bit-by-bit = Physical.

End-to-end = Transport। Source-to-destination = Network। Hop-to-hop = Data Link। Bit-by-bit = Physical।

Q42. In the OSI model, which layer(s) participate in 'end-to-end' delivery? (1) Network (2) Transport (3) Session (4) Data link

OSI model में 'end-to-end' delivery में कौन-सी layer(s) भाग लेती हैं? (1) Network (2) Transport (3) Session (4) Data link

(A) 1, 2 only **(B)** 2 only
(C) 1, 2, 3 **(D)** 2, 3, 4

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) 2 only** — In OSI terminology 'end-to-end' means between the two end TRANSPORT entities — that is the Transport layer (2 only). Network is source-to-destination; Data Link is hop-to-hop.

OSI terminology में 'end-to-end' का अर्थ दोनों end TRANSPORT entities के बीच है — यानी सिर्फ Transport layer (2 only)।

★ **Nukta:** End-to-end = Transport only. (Network = host-to-host, Data Link = hop-to-hop.)

End-to-end = सिर्फ Transport।

Q43. Which layer performs 'service point addressing'?

'Service point addressing' कौन-सी layer करती है?

(A) Network **(B)** Data Link
(C) Transport **(D)** Session

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (C) Transport** — Service point addressing = using PORT addresses to reach the correct process on a host — a Transport layer function.

Service point addressing = host पर सही process तक पहुँचने के लिए PORT addresses का उपयोग — Transport layer का काम।

★ **Nukta:** Service point = Port. Port addressing = Transport (16-bit port numbers).

Service point = Port। Port addressing = Transport (16-bit port numbers)।

Q44. Which OSI layer handles message boundaries (knowing where one message ends and the next begins)?

कौन-सी OSI layer message boundaries संभालती है (एक message कहाँ समाप्त होता है और अगला कहाँ शुरू)?

(A) Presentation **(B)** Session
(C) Transport **(D)** Network

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (C) Transport** — Message boundaries are a Transport concern: message-oriented delivery (e.g. UDP/user datagrams) preserves each message as a separate unit at the destination.

Message boundaries Transport की चिंता है: message-oriented delivery (जैसे UDP/user datagrams) हर message को अलग unit रूप में रखती है।

★ **Nukta:** Message boundary = transport-level concept (UDP message = poora message).

Message boundary = transport-level concept।

Q45. The reliable data transfer is implemented by:

Reliable data transfer कौन-सी layer implement करती है?

(A) Network layer **(B)** Physical layer
(C) Presentation layer **(D)** Transport layer

[Source: RPSC Programmer 2024 Paper-1 (with official key)]

✔ **Correct: (D) Transport layer** — Reliability (acknowledgements, retransmission, in-order delivery) is implemented by the Transport layer (e.g. TCP). (Matches the official RPSC key.)

Reliability (acknowledgements, retransmission, in-order delivery) Transport layer implement करती है (जैसे TCP)। (Official RPSC key से match.)

★ **Nukta:** Reliable delivery = Transport (TCP). IP (Network) unreliable hai — best-effort only.

Reliable delivery = Transport (TCP)। IP (Network) unreliable है।

Q46. Which layer is responsible for reliable delivery in the OSI model?

OSI model में reliable delivery के लिए कौन-सी layer जिम्मेदार है?

(A) Session **(B)** Transport
(C) Network **(D)** Application

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)]

✔ **Correct: (B) Transport** — The Transport layer ensures reliable (error-free, in-order) message delivery between processes.

Transport layer processes के बीच reliable (error-free, in-order) message delivery सुनिश्चित करती है।

★ **Nukta:** Reliable = Transport. (OSI Transport class 4 = error recovery wali service.)

Reliable = Transport।

Q47. Which layer is responsible for process-to-process delivery?

Process-to-process delivery के लिए कौन-सी layer जिम्मेदार है?

(A) Network layer **(B)** Transport layer
(C) Session layer **(D)** Data link layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (B) Transport layer** — Process-to-process (end-to-end) delivery of the entire message is the Transport layer's responsibility, using port addresses.

पूरे message की process-to-process (end-to-end) delivery Transport layer की जिम्मेदारी है, port addresses के साथ।

★ **Nukta:** Process-to-process = Transport (ports ke saath). Host-to-host = Network (IP ke saath).

Process-to-process = Transport। Host-to-host = Network।

Q48. If a port number is 16 bits (2 bytes), what is the MINIMUM size of the transport-layer header needed to carry the port addresses?

यदि port number 16 bits (2 bytes) का है, तो port addresses ले जाने के लिए transport-layer header की MINIMUM size कितनी होगी?

(A) 2 bytes **(B)** 4 bytes
(C) 8 bytes **(D)** 20 bytes

[Source: Forouzan Ch1 (Q1-23) — Practice Set]

✔ **Correct: (B) 4 bytes** — A transport header must carry at least two port numbers (source + destination) = 2 × 2 bytes = 4 bytes. (Real protocols add more: UDP header = 8 bytes, TCP header = 20 bytes.)

Transport header में कम से कम दो port numbers (source + destination) होते हैं = 2 × 2 bytes = 4 bytes। (असली protocols में और fields होते हैं: UDP = 8 bytes, TCP = 20 bytes।)

★ **Nukta:** Min transport header (ports only) = 4 bytes. UDP = 8, TCP = 20 — teeno numbers exam mein aate hain.

Min transport header (ports only) = 4 bytes। UDP = 8, TCP = 20।

Q49. Determine the OSI layer responsible for dialogue control (half-duplex or full-duplex communication).

Dialogue control (half-duplex या full-duplex communication) के लिए कौन-सी OSI layer जिम्मेदार है?

(A) Session Layer **(B)** Transport Layer
(C) Network Layer **(D)** Data Link Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (A) Session Layer** — Dialog control — deciding whose turn it is to speak (half/full-duplex dialog) — is a Session layer service.

Dialog control — किसकी बारी है (half/full-duplex dialog) — Session layer की service है।

★ **Nukta:** Dialog control + Token management + Synchronization = Session layer ka kaam.

Dialog control + Token management + Synchronization = Session layer का काम।

Q50. Which layer supports 'checkpointing' (synchronisation for recovery)?

कौन-सी layer 'checkpointing' (recovery के लिए synchronization) support करती है?

(A) Transport **(B)** Application
(C) Session **(D)** Data link

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (C) Session** — The Session layer inserts checkpoints/synchronisation points in the data stream so a dialog can resume after a failure.

Session layer data stream में checkpoints/synchronisation points लगाती है ताकि failure के बाद dialog जारी रह सके।

★ **Nukta:** Checkpointing = Session (synchronization). Ye exam ka classic sawaal hai.

Checkpointing = Session।

Q51. Which layer translates data from a format used by the application into a common format (e.g., ASCII to EBCDIC)?

कौन-सी layer data को application के format से common format में बदलती है (जैसे ASCII से EBCDIC)?

(A) Application Layer **(B)** Presentation Layer
(C) Session Layer **(D)** Transport Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (B) Presentation Layer** — Translation between data formats (ASCII/EBCDIC), encryption and compression are Presentation layer jobs.

Data formats के बीच translation (ASCII/EBCDIC), encryption और compression Presentation layer का काम है।

★ **Nukta:** Presentation = Translation + Encryption + Compression (TEC). ASCII<->EBCDIC isi layer ka classic example hai.

Presentation = Translation + Encryption + Compression।

Q52. Which layer is responsible for encryption?

Encryption के लिए कौन-सी layer जिम्मेदार है?

(A) Session **(B)** Presentation
(C) Application **(D)** Transport

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (B) Presentation** — Encryption/decryption of data for presentation is a Presentation layer responsibility (layer 6).

Presentation के लिए data की encryption/decryption Presentation layer (layer 6) की जिम्मेदारी है।

★ **Nukta:** Encryption + Compression + Translation = Presentation (L6).

Encryption + Compression + Translation = Presentation (L6)।

Q53. Which OSI layer ensures interoperability between different data formats?

कौन-सी OSI layer अलग-अलग data formats के बीच interoperability सुनिश्चित करती है?

(A) Presentation **(B)** Application
(C) Session **(D)** Network

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (A) Presentation** — The Presentation layer provides a common data representation/translation so different systems understand each other's formats.

Presentation layer एक common data representation/ translation देती है ताकि अलग systems एक-दूसरे के formats समझ सकें।

★ **Nukta:** Common format = Presentation. Isi liye ise 'syntax layer' bhi kehte hain.

Common format = Presentation। इसे 'syntax layer' भी कहते हैं।

Q54. Which OSI layer decides the type of compression?

कौन-सी OSI layer compression का प्रकार तय करती है?

(A) Application **(B)** Presentation
(C) Transport **(D)** Data link

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (B) Presentation** — Data compression (and its type) is a Presentation layer function.

Data compression (और उसका प्रकार) Presentation layer का काम है।

★ **Nukta:** Compression = Presentation. (Encryption aur translation bhi isi ke saath.)

Compression = Presentation।

Q55. Which OSI layer is responsible for encryption and compression?

Encryption और compression के लिए कौन-सी OSI layer जिम्मेदार है?

(A) Presentation Layer **(B)** Session Layer
(C) Transport Layer **(D)** Network Layer

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)]

✔ **Correct: (A) Presentation Layer** — Both encryption and compression are Presentation layer functions (layer 6).

Encryption और compression दोनों Presentation layer (layer 6) के functions हैं।

★ **Nukta:** Trio yaad rakho: Translation, Encryption, Compression = Presentation.

Trio: Translation, Encryption, Compression = Presentation।

Q56. Which layer helps to present data in one form on a host to another host in its native representation?

कौन-सी layer एक host पर किसी form में data को दूसरे host पर उसकी native representation में प्रस्तुत करने में मदद करती है?

(A) Application Layer **(B)** Presentation Layer
(C) Session Layer **(D)** Transport Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (B) Presentation Layer** — Converting between a host's internal representation and a common format is the Presentation layer's translation job.

Host की internal representation और common format के बीच बदलना Presentation layer का translation काम है।

★ **Nukta:** Native format <-> common format = Presentation (translation).

Native format <-> common format = Presentation।

Q57. _____ layer defines how data is formatted, presented, encoded, and converted for use on the network.

_____ layer तय करती है कि data network में उपयोग के लिए कैसे format, present, encode और convert होता है।

(A) Transport **(B)** Presentation
(C) Application **(D)** Physical

[Source: Central exams — SSC/Bank/Insurance Computer Awareness (BYJU'S)]

✔ **Correct: (B) Presentation** — This is the textbook definition of the Presentation layer — formatting, encoding and conversion of data.

यह Presentation layer की textbook definition है — data का formatting, encoding और conversion।

★ **Nukta:** 'Formatted, presented, encoded, converted' = Presentation ke liye mnemonic-type line.

'Formatted, presented, encoded, converted' = Presentation।

Q58. Which layer provides services to the user?

कौन-सी layer user को services देती है?

(A) Application layer **(B)** Session layer
(C) Presentation layer **(D)** Physical layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) Application layer** — The Application layer (layer 7) provides network services directly to user applications (HTTP, FTP, SMTP, DNS...).

Application layer (layer 7) user applications को सीधे network services देती है (HTTP, FTP, SMTP, DNS...)।

★ **Nukta:** User ke sabse kareeb = Application layer (L7).

User के सबसे करीब = Application layer (L7)।

Q59. A server at a company is using TCP/IP to connect to the network and is running multiple applications. At which OSI layer is this server operating?

एक company का server network से जुड़ने के लिए TCP/IP use कर रहा है और अनेक applications चला रहा है। यह server किस OSI layer पर काम कर रहा है?

(A) Application Layer **(B)** Physical Layer
(C) Network Layer **(D)** Transport Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) Application Layer** — Running multiple applications that serve users = Application layer work (the applications use the lower layers for delivery).

users के लिए अनेक applications चलाना = Application layer का काम (applications नीचे की layers use करती हैं)।

★ **Nukta:** Server + running applications = Application layer.

Server + applications = Application layer।

Q60. Simple Network Management Protocol (SNMP) runs on:

Simple Network Management Protocol (SNMP) किस layer पर चलता है?

(A) Application layer **(B)** Transport layer
(C) Network layer **(D)** Data link layer

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (A) Application layer** — SNMP is an Application layer protocol (it uses UDP at the transport layer to manage network devices).

SNMP एक Application layer protocol है (network devices manage करने के लिए यह UDP use करता है)।

★ **Nukta:** SNMP/DNS/HTTP/FTP/SMTP = Application layer protocols. UDP/TCP = Transport. IP = Network.

SNMP/DNS/HTTP/FTP/SMTP = Application layer protocols।

Q61. Application layer protocol defines:
Application layer protocol क्या define करता है?
(A) Types of messages exchanged / कौन-से messages आदान-प्रदान होते हैं **(B)** Rules for when and how processes send and respond to messages / processes messages कब और कैसे भेजें/जवाब दें — इसके rules
(C) Message format, syntax and semantics / Message format, syntax और semantics **(D)** All of the above / उपरोक्त सभी
[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (D) All of the above** — An application-layer protocol defines the message types, the rules for when/how processes send and respond, and the message format (syntax + semantics) — all of the above.
Application-layer protocol message types, भेजने/जवाब के rules, और message format (syntax + semantics) — सब define करता है।
★ **Nukta:** Application protocol = Message types + Rules + Format. 'All of the above' hi jawab hai.
Application protocol = Message types + Rules + Format।

Q62. SMTP, FTP and DNS are the protocols of which layer of the OSI model?
SMTP, FTP और DNS OSI model की किस layer के protocols हैं?
(A) Network layer **(B)** Transport layer
(C) Application layer **(D)** Data Link layer
[Source: Central exams — SSC/Railway/UPSC Computer Awareness (Exampundit)]

✔ **Correct: (C) Application layer** — SMTP (e-mail), FTP (file transfer) and DNS (name resolution) are all Application layer protocols (layer 7).
SMTP (e-mail), FTP (file transfer) और DNS (name resolution) — तीनों Application layer protocols (layer 7) हैं।
★ **Nukta:** User-facing protocols (HTTP, FTP, SMTP, DNS, POP3, IMAP) = Application layer.
User-facing protocols = Application layer।

Q63. Consider the statements: S1: Network layer handles both logical addressing and routing. S2: Transport layer handles segmentation and reassembly. Which is correct?
Statements देखिए: S1: Network layer logical addressing और routing दोनों संभालती है। S2: Transport layer segmentation और reassembly संभालती है। कौन सा सही है?
(A) Only S1 **(B)** Only S2
(C) Both S1 and S2 **(D)** Neither S1 nor S2
[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (C) Both S1 and S2** — Both statements are textbook-correct: Network = logical addressing + routing; Transport = segmentation + reassembly.
दोनों statements textbook-correct हैं: Network = logical addressing + routing; Transport = segmentation + reassembly।
★ **Nukta:** Network = IP addressing + Routing. Transport = Segmentation + Reassembly. Dono sahi.
Network = IP addressing + Routing। Transport = Segmentation + Reassembly।

Section C — TCP/IP Model

TCP/IP (DoD, 1970s): 4/5 layers, layer mapping to OSI, Internet layer protocols, address levels.

Q64. TCP/IP model does not have _____ layer but the OSI model has it.
TCP/IP model में _____ layer नहीं है पर OSI model में है।
(A) Session layer **(B)** Presentation layer
(C) Application layer **(D)** Both (a) and (b)
[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (D) Both (a) and (b)** — TCP/IP has no separate Session or Presentation layers — their functions are absorbed by the TCP/IP Application layer. (OSI = 7 layers; TCP/IP = 4 layers.)
TCP/IP में Session या Presentation की अलग layer नहीं है — उनके काम TCP/IP की Application layer में आ जाते हैं।
★ **Nukta:** TCP/IP mein Session + Presentation nahi hoti — dono Application mein merge ho jaati hain.
TCP/IP में Session + Presentation नहीं होती — दोनों Application में merge होती हैं।

Q65. The TCP/IP model was developed _____ the OSI model.
TCP/IP model, OSI model के _____ develop हुआ।
(A) prior to **(B)** after
(C) simultaneous to **(D)** none of these
[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) prior to** — TCP/IP was developed in the 1970s (by Cerf & Kahn for ARPANET/DoD) — BEFORE the OSI model (published by ISO in 1984).
TCP/IP 1970s में develop हुआ (Cerf & Kahn ने ARPANET/DoD के लिए) — OSI model (1984) से पहले।
★ **Nukta:** TCP/IP (1970s, adopted 1983) PEHLE; OSI (1984) BAAD mein. 'Prior to' yaad rakho.
TCP/IP (1970s) पहले; OSI (1984) बाद में।

Q66. The TCP/IP _____ layer is equivalent to the combined Session, Presentation and Application layers of the OSI model.

TCP/IP की _____ layer, OSI model की Session, Presentation और Application layers के समकक्ष है।

(A) application **(B)** network
(C) data link **(D)** physical

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) application** — The TCP/IP Application layer covers what OSI layers 5 + 6 + 7 do (application + presentation + session).

TCP/IP Application layer वही करती है जो OSI की layers 5 + 6 + 7 करती हैं।

★ **Nukta:** Mapping: TCP/IP Application = OSI Session+Presentation+Application; TCP/IP Internet = OSI Network; TCP/IP Network Access = OSI Data Link+Physical.

Mapping: TCP/IP Application = OSI 5+6+7; Internet = Network; Network Access = Data Link+Physical।

Q67. Which of the following is NOT a part of the TCP/IP model?

निम्नलिखित में से कौन TCP/IP model का हिस्सा नहीं है?

(A) Application **(B)** Transport
(C) Data link **(D)** Session

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)]

✔ **Correct: (D) Session** — TCP/IP (4-layer) = Application, Transport, Internet, Network Access. Session is an OSI layer with no separate TCP/IP layer.

TCP/IP (4-layer) = Application, Transport, Internet, Network Access। Session OSI की layer है, TCP/IP में अलग नहीं।

★ **Nukta:** TCP/IP ke 4 layers: Application, Transport, Internet, Network Access. Session sirf OSI mein.

TCP/IP की 4 layers: Application, Transport, Internet, Network Access। Session सिर्फ OSI में।

Q68. The number of layers in the Internet protocol stack (reference model) is:

Internet protocol stack (reference model) में layers की संख्या कितनी है?

(A) 5 **(B)** 7
(C) 6 **(D)** 3

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) 5** — The TCP/IP Internet reference (protocol) stack has 5 layers: Physical, Data Link, Network, Transport, Application. (The original TCP/IP suite has 4 — Network Access merges Physical+Data Link.)

TCP/IP Internet reference stack में 5 layers हैं: Physical, Data Link, Network, Transport, Application। (मूल TCP/IP suite में 4 — Network Access में Physical+Data Link मिले हुए।)

★ **Nukta:** 5-layer Internet stack vs 4-layer TCP/IP suite vs 7-layer OSI — teeno numbers exam mein aate hain!

5-layer Internet stack, 4-layer TCP/IP suite, 7-layer OSI — तीनों numbers याद रखो!

Q69. How many layers are there in the TCP/IP model (original 4-layer suite)?

TCP/IP model (मूल 4-layer suite) में कितनी layers हैं?

(A) Seven **(B)** Five
(C) Four **(D)** Nine

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)]

✔ **Correct: (C) Four** — The original TCP/IP model has 4 layers: Network Access, Internet, Host-to-Host (Transport), Application.

मूल TCP/IP model में 4 layers हैं: Network Access, Internet, Host-to-Host (Transport), Application।

★ **Nukta:** 4-layer TCP/IP: Network Access + Internet + Transport (Host-to-Host) + Application.

4-layer TCP/IP: Network Access + Internet + Transport + Application।

Q70. Which of the following is NOT a layer of the TCP/IP model? (Host-to-Host / Internet / Network Access / Physical)

निम्नलिखित में से कौन TCP/IP model की layer नहीं है? (Host-to-Host / Internet / Network Access / Physical)

(A) Host to Host **(B)** Internet
(C) Network Access **(D)** Physical

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)]

✔ **Correct: (D) Physical** — The 4-layer TCP/IP suite has no separate Physical layer — physical functions are part of Network Access.

4-layer TCP/IP suite में अलग Physical layer नहीं है — physical functions Network Access का हिस्सा हैं।

★ **Nukta:** TCP/IP ki 4 layers: Network Access, Internet, Host-to-Host, Application. 'Physical' alag layer nahi hai.

TCP/IP की 4 layers: Network Access, Internet, Host-to-Host, Application।

Q71. The Data Link layer and Physical layer of the OSI model combine to form the _____ layer of the TCP/IP model.

OSI model की Data Link और Physical layers मिलकर TCP/IP model की _____ layer बनाती हैं।

(A) Network Access **(B)** Internet
(C) Host to Host **(D)** Application

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)]

✔ **Correct: (A) Network Access** — OSI layers 1 + 2 (Physical + Data Link) map to the single TCP/IP Network Access layer.

OSI की layers 1 + 2 (Physical + Data Link) मिलकर TCP/IP की Network Access layer बनाती हैं।

★ **Nukta:** OSI 1+2 = TCP/IP Network Access. OSI 5+6+7 = TCP/IP Application. Bas ye 2 mappings yaad rakho.

OSI 1+2 = Network Access। OSI 5+6+7 = Application।

Q72. IP (Internet Protocol) is a protocol of the _____ layer of the TCP/IP model.

IP (Internet Protocol) TCP/IP model की _____ layer का protocol है.

(A) Network Access **(B)** Internet
(C) Transport **(D)** Application

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)]

✔ **Correct: (B) Internet** — IP, ICMP, ARP work at the Internet layer of TCP/IP (equivalent to the OSI Network layer).

IP, ICMP, ARP TCP/IP की Internet layer पर काम करते हैं (OSI Network layer के समकक्ष)।

★ **Nukta:** IP = Internet layer. TCP/UDP = Transport. Ethernet = Network Access.

IP = Internet layer। TCP/UDP = Transport। Ethernet = Network Access।

Q73. In TCP/IP, which layer uses specific (user-friendly) addresses — e.g. a URL like example.com?

TCP/IP में कौन-सी layer specific (user-friendly) addresses use करती है — जैसे example.com जैसी URL?

(A) Application layer **(B)** Network layer
(C) Data link layer **(D)** Physical layer

[Source: Forouzan Ch1 (Q1-24) — Practice Set]

✔ **Correct: (A) Application layer** — Forouzan's four address levels: specific/user-friendly addresses (URLs, domain names) are used at the Application layer; port addresses at Transport; IP at Network; physical (MAC) at Data Link.

Forouzan के चार address levels: specific/user-friendly addresses (URL, domain name) Application layer पर; port addresses Transport पर; IP Network पर; MAC Data Link पर।

★ **Nukta:** Address levels: Application = specific (URL), Transport = port (16-bit), Network = IP (32/128-bit), Data Link = MAC (48-bit).

Address levels: Application = specific, Transport = port, Network = IP, Data Link = MAC।

Q74. In the TCP/IP model, a link-layer switch operates at which layers?

TCP/IP model में link-layer switch किन layers पर काम करता है?

(A) Network and Transport layers **(B)** Physical and Data Link layers
(C) Application layer only **(D)** All five layers

[Source: Forouzan Ch1 (Q1-16) — Practice Set]

✔ **Correct: (B) Physical and Data Link layers** — A link-layer switch reads MAC addresses and regenerates signals — it works at the Physical and Data Link layers (the TCP/IP Network Access layer).

link-layer switch MAC addresses पढ़ता है और signals regenerate करता है — यह Physical और Data Link layers (Network Access) पर काम करता है।

★ **Nukta:** Switch = L1+L2 (Network Access). Router = L3 tak. Gateway = L4+.

Switch = L1+L2। Router = L3 तक। Gateway = L4+।

Section D — Comparison, Encapsulation & PDUs

OSI vs TCP/IP, encapsulation order & direction, PDUs per layer, numericals.

Q75. Identify the correct order of data encapsulation from Layer 7 to Layer 1:

Layer 7 से Layer 1 तक data encapsulation का सही क्रम पहचानिए:

(A) Data → Segment → Packet → Frame → Bits **(B)** Data → Packet → Segment → Frame → Bits
(C) Bits → Frame → Packet → Segment → Data **(D)** Data → Frame → Packet → Segment → Bits

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)]

✔ **Correct: (A) Data → Segment → Packet → Frame → Bits** — Going down the OSI stack: Application data → Transport segment → Network packet → Data Link frame → Physical bits.

OSI stack में नीचे जाते हुए: Application data → Transport segment → Network packet → Data Link frame → Physical bits।

★ **Nukta:** Encapsulation order: Data -> Segment -> Packet -> Frame -> Bits. Mnemonic: 'Do Some People Fear Barking?'

Encapsulation: Data -> Segment -> Packet -> Frame -> Bits।

Q76. Encapsulation happens from:

Encapsulation किस दिशा में होती है?

(A) Top to bottom **(B)** Bottom to top
(C) Random **(D)** Parallel

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)]

✔ **Correct: (A) Top to bottom** — At the sender, each layer adds its header going DOWN the stack (top to bottom); the receiver removes headers going UP (bottom to top).

Sender पर हर layer नीचे जाते हुए अपना header जोड़ती है (top to bottom); receiver पर ऊपर जाते हुए headers हटते हैं (bottom to top)।

★ **Nukta:** Sender = encapsulation (top->bottom, headers add). Receiver = de-encapsulation (bottom->top, headers remove).

Sender = encapsulation (top->bottom)। Receiver = de-encapsulation (bottom->top)।

Q77. In the layer hierarchy, as the data packet moves from the upper to the lower layers, headers are:

Layer hierarchy में, जब data packet ऊपर से नीचे की layers की ओर जाता है, तो headers:

(A) Added **(B)** Removed
(C) Rearranged **(D)** Modified

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)]

✔ **Correct: (A) Added** — Each lower layer ADDS its own header (and sometimes a trailer) to the data unit received from above — that is encapsulation.

हर नीचे की layer ऊपर से आए data unit में अपना header (कभी-कभी trailer) जोड़ती है — यही encapsulation है।

★ **Nukta:** Down the stack = headers ADD. Up the stack = headers REMOVE.

नीचे की ओर = headers ADD। ऊपर की ओर = headers REMOVE।

Q78. A source S and destination D are connected through ONE intermediate router R. How many times does a packet visit the Network layer and the Data Link layer during the journey?

Source S और destination D के बीच एक intermediate router R है। यात्रा के दौरान packet Network layer और Data Link layer को कितनी बार visit करता है?

(A) Network layer - 4 times, Data link layer - 4 times
(B) Network layer - 4 times, Data link layer - 6 times
(C) Network layer - 2 times, Data link layer - 4 times
(D) Network layer - 3 times, Data link layer - 4 times

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)]

✔ **Correct: (D) Network layer - 3 times, Data link layer - 4 times** — Network layer: once at S (send), once at R (forward), once at D (receive) = 3. Data Link: 2 links (S→R, R→D), each visited twice (send + receive) = 4.

Network layer: S पर 1, R पर 1, D पर 1 = 3 बार। Data Link: 2 links (S→R, R→D), हर link 2 बार (send + receive) = 4 बार।

★ **Nukta:** Formula: n routers -> Network layer visits = n+2, Data Link visits = 2(n+1). Yahan n=1 -> NL=3, DL=4.

Formula: n routers -> Network layer = n+2, Data Link = 2(n+1)। यहाँ n=1 -> NL=3, DL=4।

Q79. In TCP/IP, which data unit is encapsulated INSIDE a frame?

TCP/IP में कौन-सा data unit frame के अंदर encapsulate होता है?

(A) A segment **(B)** A datagram
(C) A bit **(D)** A message

[Source: Forouzan Ch1 (Q1-20) — Practice Set]

✔ **Correct: (B) A datagram** — The Data Link layer encapsulates the Network layer's packet (datagram) inside a frame (frame = header + datagram + trailer).

Data Link layer Network layer के packet (datagram) को frame के अंदर encapsulate करती है (frame = header + datagram + trailer)।

★ **Nukta:** Frame ke andar DATAGRAM hota hai. Segment -> Datagram -> Frame -> Bits.

Frame के अंदर DATAGRAM होता है।

Q80. In the TCP/IP model, the unit of data at the transport layer is called a:

TCP/IP model में transport layer पर data की unit कहलाती है:

(A) Packet **(B)** Segment
(C) Frame **(D)** Stream

[Source: Forouzan Ch1 (Q1-20/21) — Practice Set]

✔ **Correct: (B) Segment** — Transport layer data unit = segment (TCP) or user datagram (UDP); Network = datagram/packet; Data Link = frame; Physical = bits.

Transport layer की data unit = segment (TCP) या user datagram (UDP); Network = datagram/packet; Data Link = frame; Physical = bits।

★ **Nukta:** PDU pairs: Transport = Segment, Network = Datagram/Packet, Data Link = Frame, Physical = Bits.

PDU: Transport = Segment, Network = Datagram, Data Link = Frame, Physical = Bits।

Q81. A system uses five protocol layers. The application creates a 100-byte message and EVERY layer (including the 5th and the 1st) adds a 10-byte header. What is the efficiency (application bytes / total transmitted bytes)?

एक system में पाँच protocol layers हैं। Application 100-byte message बनाती है और हर layer (5वीं और 1वीं सहित) 10-byte header जोड़ती है। Efficiency कितनी होगी (application bytes / कुल transmitted bytes)?

(A) 50% **(B)** 66.7%
(C) 75% **(D)** 80%

[Source: Forouzan Ch1 (P1-11) — Practice Set (Problem)]

✔ **Correct: (B) 66.7%** — Total bytes = 100 (message) + 5 × 10 (headers) = 150. Efficiency = 100/150 = 2/3 ≈ 66.7%.

कुल bytes = 100 (message) + 5 × 10 (headers) = 150।
Efficiency = 100/150 = 2/3 ≈ 66.7%।

★ **Nukta:** Encapsulation overhead numerical: efficiency = data / (data + headers). 100/(100+50) = 66.7%.

Efficiency = data / (data + headers) = 100/150 = 66.7%।

★ Nukte Rapid-Fire — Exam Pointers (quick revision)

One-line question (left) with its exam answer + trick (right). Revise these before the exam.

OSI layers mnemonic? OSI layers की mnemonic?	✔ Please Do Not Throw Sausage Pizza Away (L1→L7) Please Do Not Throw Sausage Pizza Away (L1→L7)
OSI kisne banaya, kab? OSI किसने बनाया, कब?	✔ ISO, in 1984 — 7 layers ISO, 1984 में — 7 layers
TCP/IP kisne banaya? TCP/IP किसने बनाया?	✔ Cerf & Kahn (DoD/ARPA), 1970s — adopted 1983 Cerf & Kahn (DoD/ARPA), 1970s — adopted 1983
Layer counts? Layer counts?	✔ OSI = 7, Internet stack = 5, TCP/IP suite = 4 OSI = 7, Internet stack = 5, TCP/IP suite = 4
TCP/IP mein kya missing? TCP/IP में क्या missing?	✔ No separate Session & Presentation layers Session और Presentation की अलग layers नहीं
OSI→TCP/IP mapping? OSI→TCP/IP mapping?	✔ OSI 1+2 = Network Access; OSI 5+6+7 = Application OSI 1+2 = Network Access; OSI 5+6+7 = Application
PDU per layer? हर layer की PDU?	✔ Data / Segment / Packet-Datagram / Frame / Bits Data / Segment / Packet-Datagram / Frame / Bits
Encapsulation direction? Encapsulation की दिशा?	✔ Sender: top→bottom (headers add); Receiver: bottom→top (remove) Sender: top→bottom; Receiver: bottom→top

4 levels of addresses? 4 levels के addresses?	✔ Physical(48-bit), IP(32/128-bit), Port(16-bit), Specific(URL) Physical(48-bit), IP(32/128-bit), Port(16-bit), Specific(URL)
Delivery scopes? Delivery के scopes?	✔ Bit-by-bit=Physical; Hop-to-hop=Data Link; Source-destination=Network; End-to-end/Process-process=Transport Bit-by-bit=Physical; Hop-to-hop=Data Link; Source-destination=Network; End-to-end=Transport
Presentation layer kaam? Presentation layer के काम?	✔ Translation + Encryption + Compression Translation + Encryption + Compression
Session layer ke 3 kaam? Session layer के 3 काम?	✔ Dialog control + Token management + Synchronization (checkpoint) Dialog control + Token management + Synchronization
Data Link ka trio? Data Link का trio?	✔ Framing + Error detection + Physical(MAC) addressing Framing + Error detection + MAC addressing
LLC vs MAC? LLC बनाम MAC?	✔ LLC (802.2) = Network layer interface; MAC (802.3) = medium access + MAC address LLC = interface; MAC = medium access + address
Support layers grouping? Support layers grouping?	✔ Network support = 1,2,3; User support = 5,6,7; Transport(4) = bridge Network support = 1,2,3; User support = 5,6,7; Transport = bridge
Device layers? Devices की layers?	✔ Repeater/Hub=L1; Bridge/Switch=L2; Router=L3; Gateway=L4+ Repeater/Hub=L1; Bridge/Switch=L2; Router=L3; Gateway=L4+
Reliable delivery? Reliable delivery?	✔ Transport layer (TCP) — IP is best-effort only Transport layer (TCP) — IP सिर्फ best-effort
Router visits formula? Router visits का formula?	✔ n routers: Network layer = n+2, Data Link = 2(n+1) n routers: Network layer = n+2, Data Link = 2(n+1)
Physical layer ka kaam? Physical layer का काम?	✔ Signals, media, line/channel coding, modulation, baud rate, bit-by-bit Signals, media, coding, modulation, baud rate, bit-by-bit
IP/ICMP/ARP kis layer ke? IP/ICMP/ARP किस layer के?	✔ Network / Internet layer (L3) Network / Internet layer (L3)